

Restaurant Customer Analytics Platform

Solution Design Document

Version

1.0

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1. Executive Summary

The objective of this solution is to build an enterprise-grade customer analytics platform capable of providing actionable business insights across restaurant locations and ordering channels. The platform will calculate Customer Lifetime Value (CLV) daily, perform customer segmentation using Recency-Frequency-Monetary (RFM) analysis, identify churn risk indicators, measure loyalty program effectiveness, analyze sales trends and seasonality, evaluate discount effectiveness, and rank location performance.

The solution is designed entirely using AWS native services and PySpark, adhering to organizational requirements to avoid third-party platforms such as Snowflake or DBT.

2. Business Objectives

The platform will provide insights into:

- Customer Lifetime Value (CLV)
- Customer Segmentation (RFM)
- Churn Risk Identification
- Sales Trends and Seasonality
- Loyalty Program Performance
- Location Performance
- Pricing and Discount Effectiveness

Business users will access these insights through a Streamlit dashboard.

3. Source Systems

SQL Server Source Tables

order_items

Stores transactional order item data.

Key Attributes:

- order_id
- user_id
- restaurant_id
- creation_time_utc
- item_price
- item_quantity
- is_loyalty

order_item_options

Stores item modifiers, add-ons, and discounts.

Key Attributes:

- order_id
- lineitem_id
- option_price
- option_quantity

date_dim

Provides calendar attributes for reporting and seasonality analysis.

4. Architecture Overview

The architecture follows a Medallion Data Lake pattern:

Bronze Layer

Raw source ingestion.

Silver Layer

Validated and standardized business entities.

Gold Layer

Business-ready analytics tables.

5. AWS Services Used

Service	Purpose
SQL Server	Source System
AWS DMS	Incremental Data Ingestion
Amazon S3	Data Lake Storage
AWS Glue	ETL Processing
AWS Glue Data Quality	Data Validation
AWS Glue Catalog	Metadata Management
Amazon Athena	SQL Analytics
AWS CloudWatch	Monitoring
AWS SNS	Alerting
AWS Secrets Manager	Credential Management
AWS KMS	Encryption
Streamlit	Dashboarding
GitHub Actions	CI/CD

6. Data Flow

1. SQL Server tables are replicated using AWS DMS.
 2. Data is stored in the Bronze layer.
 3. AWS Glue validates incoming data.
 4. PySpark transforms data into Silver datasets.
 5. Business metrics are calculated and stored in Gold.
 6. Glue Catalog registers datasets.
 7. Athena queries Gold datasets.
 8. Streamlit dashboards consume Athena views.
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7. Data Model

Gold Tables

customer_ltv_daily

Tracks daily CLV evolution.

Columns:

- customer_id
- date_key
- daily_revenue
- cumulative_revenue
- clv_bucket

customer_rfm

Stores segmentation metrics.

Columns:

- recency_score
- frequency_score
- monetary_score
- customer_segment

customer_churn

Stores churn indicators.

Columns:

- days_since_last_order
- avg_days_between_orders
- spend_change_pct
- churn_status

sales_trends

Aggregated daily, weekly, monthly revenue.

loyalty_impact

Comparison of loyalty versus non-loyalty customers.

location_performance

Performance ranking by restaurant.

discount_analysis

Measures impact of discounts on revenue.

8. Customer Lifetime Value Logic

Net Order Revenue:

Revenue = (Item Price × Item Quantity)

+ (Option Price × Option Quantity)

Daily CLV:

Running cumulative customer revenue over time.

Customer groups:

- High CLV = Top 20%
 - Medium CLV = Middle 60%
 - Low CLV = Bottom 20%
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9. Customer Segmentation

RFM Metrics:

Recency: Days since last purchase

Frequency: Number of purchases during prior 12 months

Monetary: Total spend during prior 12 months

Segments:

- VIP
- Loyal
- New Customer
- At Risk

- Lost Customer
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10. Churn Indicators

Calculated Metrics:

- Days Since Last Order
- Average Days Between Orders
- Spend Trend %

Risk Categories:

- Active (0-30 Days)
 - Warning (31-45 Days)
 - At Risk (46-90 Days)
 - Churned (>90 Days)
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11. Security Design

Security controls include:

- AWS KMS encryption
 - IAM least privilege access
 - Secrets Manager credential storage
 - HTTPS/TLS encryption
 - S3 bucket versioning
 - CloudWatch monitoring
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12. Monitoring and Alerting

CloudWatch monitors:

- Glue Jobs
- DMS Replication
- Athena Queries

SNS sends alerts for:

- ETL failures
- Validation failures
- Data quality violations

13. CI/CD Design

GitHub Actions Pipeline:

Source Code Commit → Lint → Unit Tests → Package Deployment → Deploy Glue Jobs → Deploy Streamlit Application

14. Dashboard Requirements

The Streamlit dashboard includes:

1. Customer Segmentation
 2. Customer Lifetime Value
 3. Churn Risk Indicators
 4. Sales Trends & Seasonality
 5. Loyalty Program Impact
 6. Location Performance
 7. Pricing & Discount Effectiveness
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15. Benefits

The solution provides:

- Daily CLV visibility
- Customer retention insights
- Improved marketing targeting
- Operational performance monitoring
- Promotion effectiveness measurement
- Scalable cloud-native architecture
- Secure enterprise-grade analytics platform